THE LAWS OF SOFTWARE TEAMS

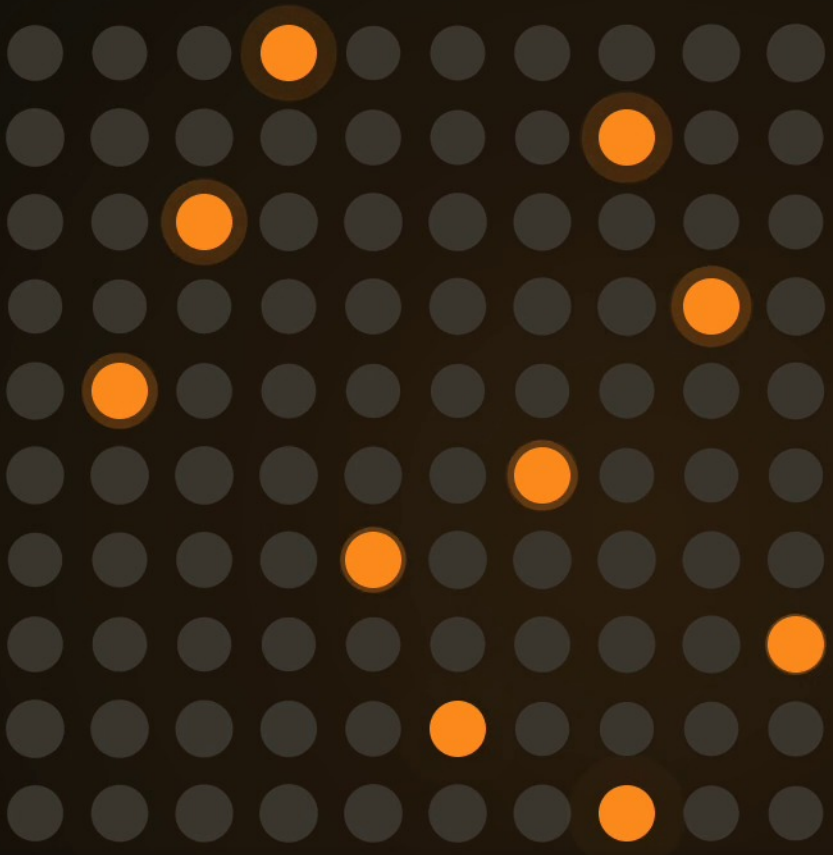
Price's law

The square root of the team does half the work

swipe →

Ivan Rykov

100 people on a team
10 of them do HALF the work



Price's law

√N people
do half of the work



Derek J. de Solla Price

Physicist & historian of science, Yale

"Little Science, Big Science", 1963

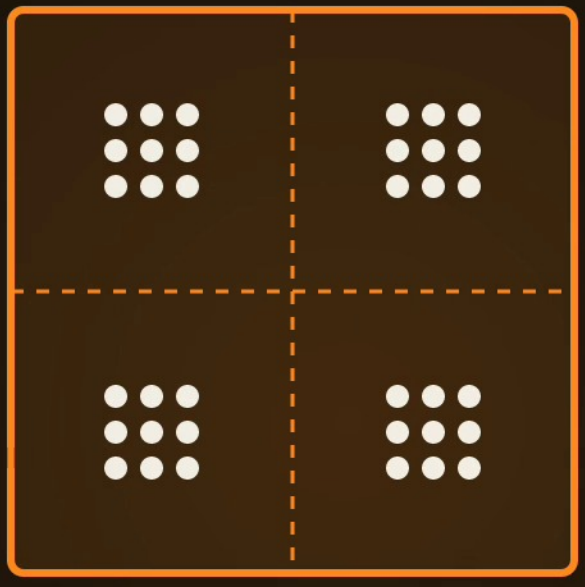
The bigger the team, the smaller the core

team size → people doing half the work

10 people → 3	30%
100 people → 10	10%
1,000 people → 32	3.2%
10,000 people → 100	1%

√N grows far slower than N

Want 2× the output? May need 4× the people



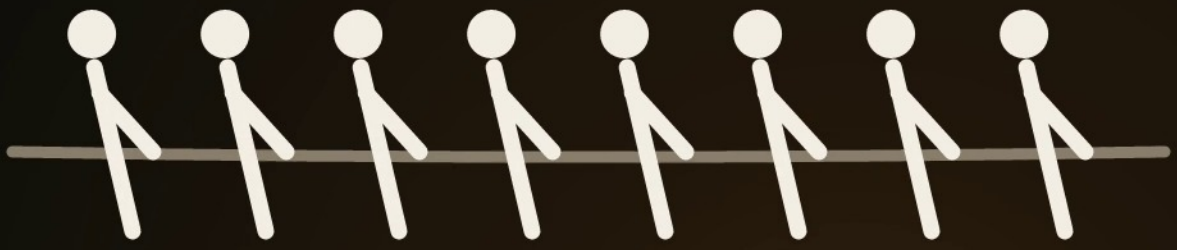
```
side = output  ×2
area = people  ×4
```

One reason:

the Ringelmann effect

1913 · pulling experiments

More people, less pull from each





8 people 49% effort each

In open source, it's even steeper

Apache web server · developers by new code



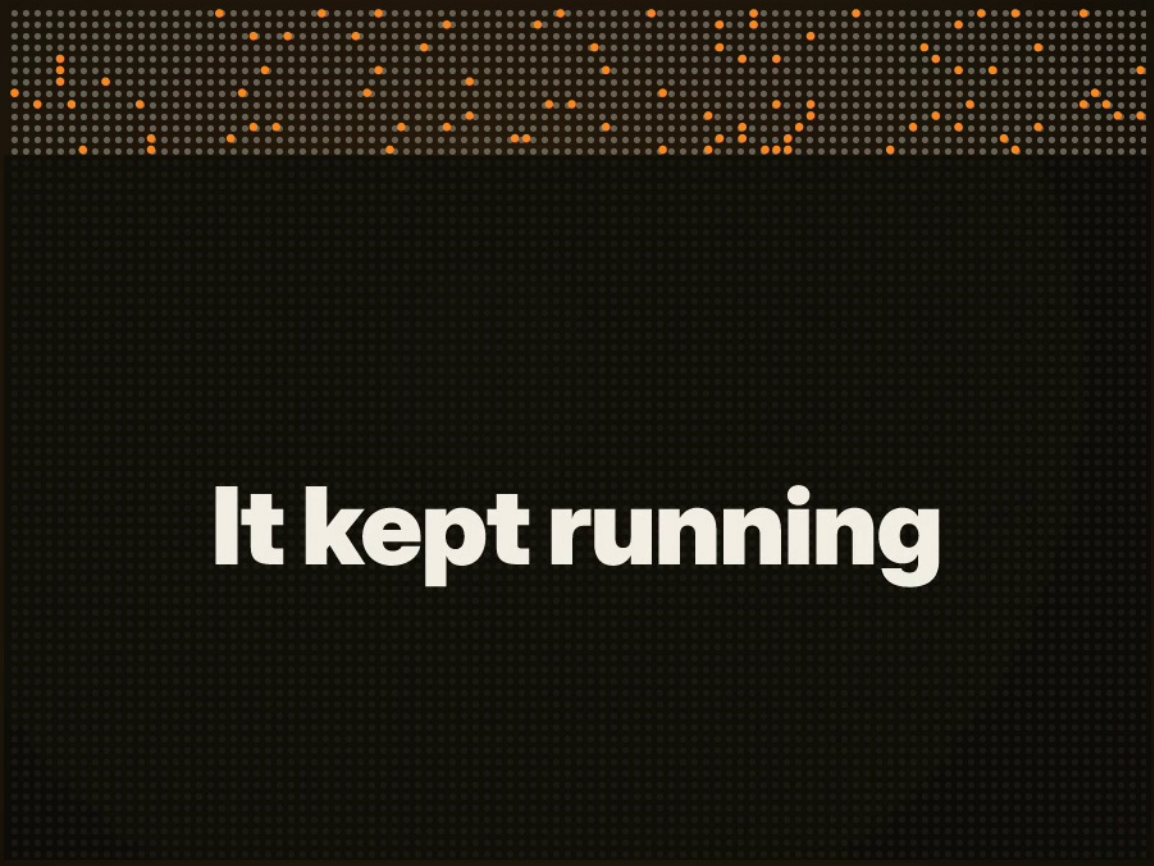
Mockus, Fielding & Herbsleb, 2002

88% of new code came from the top 15

07 CASE STUDY

Twitter, late 2022

~7,500 employees



√7,500 ≈ 87 → half the work

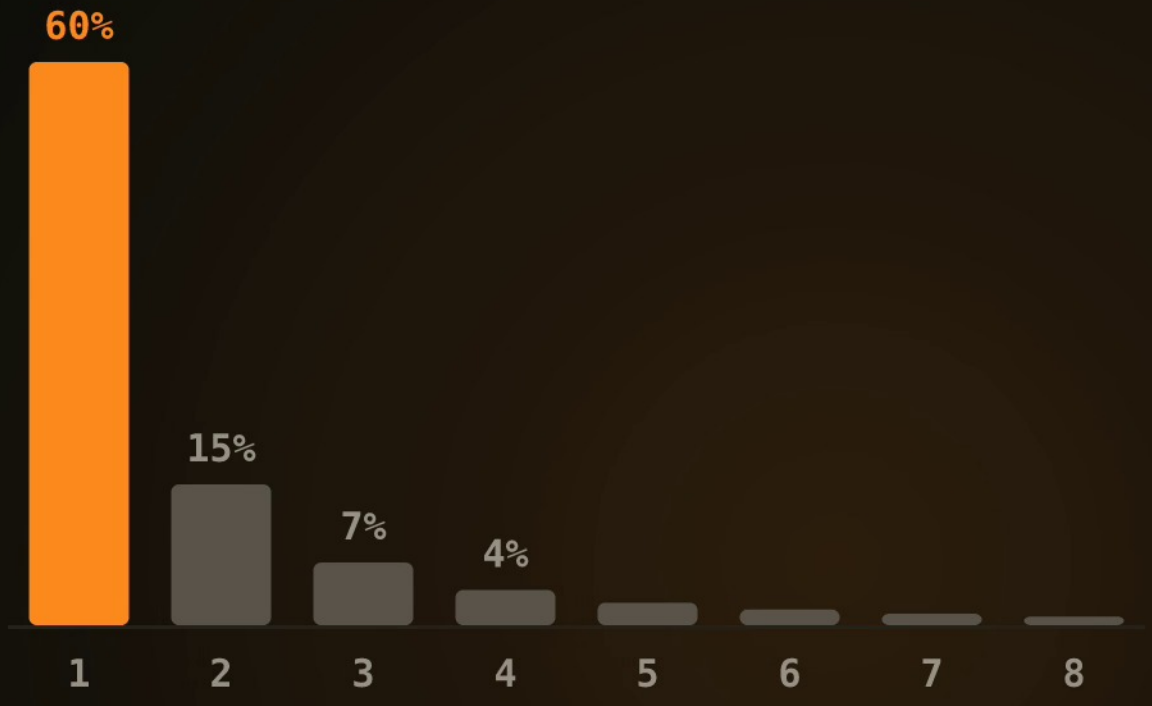
Jan 2023: ~1,300 still working

Source: CNBC

Plot twist:
it's not a real law

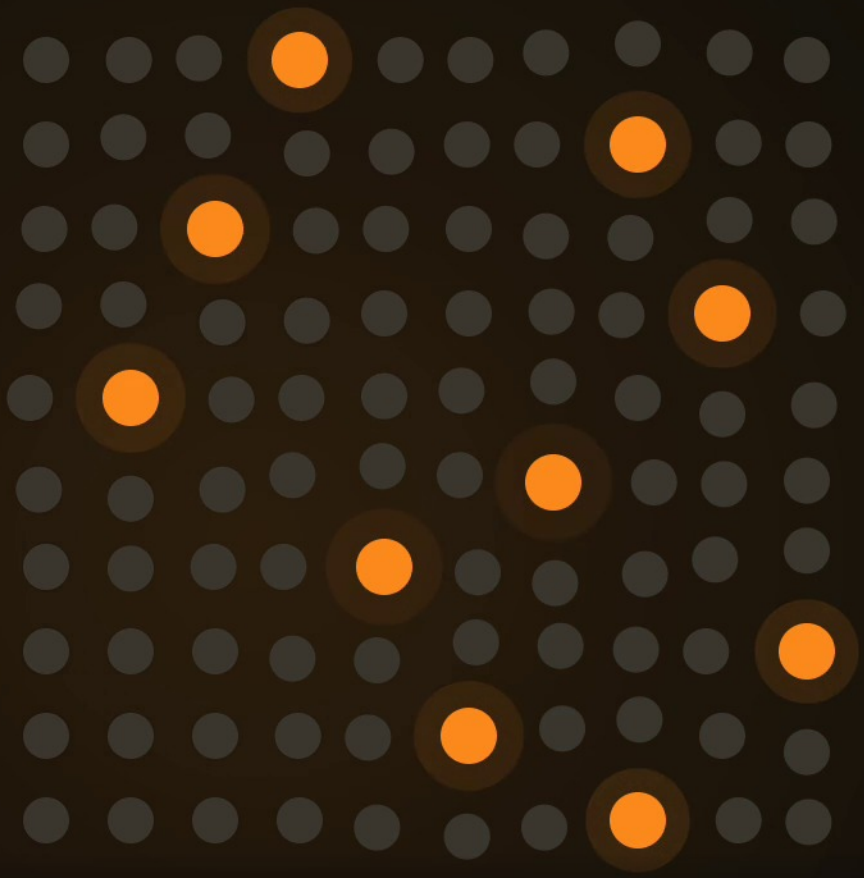
The exact numbers didn't hold, but output stays lopsided

Nicholls, 1988



Lotka, 1926: 60% of authors publish only once

So don't count heads
Find your √N
Then clear their path



WHAT I TAKE FROM THIS

This is how I work with this law:

1. I find the few strong people who do half of the work. They stand out in results and in how they act: they ask questions, suggest ideas, finish tasks early and say so, and act before problems appear
2. I clear their path: no endless approvals, and a whole area to own
3. But I still check the direction and support them, especially when they take on a new area. A fast person can build the wrong thing or build it the wrong way. The work looks fast, but the result is poor

WHAT I TAKE FROM THIS

4. Even though they own an area, nothing depends on them alone. Knowledge is written down, on-call and review duties rotate, and discussions happen in open channels
5. Strong people get tired too. I limit their parallel tasks, and weekend work is an agreed exception with a day off after it
6. For everyone else, my approach depends on their potential and level. Where someone could do more, I create the conditions: their own sub-product, more responsibility or a small working group inside the team, with a clear part for each person

WHAT I TAKE FROM THIS

AI makes the gap between people even wider: people who use it well become much more effective than others. But with AI, some finish early and do not say they are done. Whether their real speed shows depends on the manager, the task, code reviews and tests, and how strong the team is

Price's law is not a reason to keep the top few and let the rest go. I keep it in mind when I form and split teams, choose their size and plan the org structure

Sources: Derek de Solla Price, 1963 · Lotka, 1926 · CNBC, 2023 · Mockus, Fielding and Herbsleb, 2002 · Ringelmann, 1913 · Nicholls, 1988

next: Conway's law